NetSec_RepeatCA_Assignment - B00141800

introduction

This project aimed to create a secure multi-site network using Cisco Packet Tracer, connecting Dublin, London, and Beijing sites. To achieve this, various security measures were put in place to protect the network, control access, and enhance monitoring. The goal was to design a network that could safeguard against potential threats while allowing authorised users to access the system. By implementing different security protocols, the network's vulnerability to attacks was reduced, and its overall security was improved. This was done to ensure the confidentiality, integrity, and availability of the network.

During the project, I implemented several security features including device hardening, SSH, AAA authentication, access control lists (ACLs), SYSLOG monitoring, OSPF authentication, NAT/PAT, port security, DHCP snooping and wireless security. These features were tested where possible to confirm that they were operating correctly.

The goal was not just to get the network up and running, but also to make it more secure. This meant stopping people from accessing the network without permission, controlling how different parts of the network talk to each other, and keeping the devices on the network safe.

2. Device hardening

2.1 password security

To better secure the Dublin router, I started by setting up a key security feature: password protection. This involved creating an enable secret to lock down privileged EXEC mode, which is a critical part of the router's configuration. I also made sure to turn on password encryption, so that when someone looks at the router's settings, they won't see the passwords in plain text. This helps keep the router safe from unauthorised access.

The enable secret is a key security feature that stops unauthorised users from getting into the router's admin settings. It's like an extra lock on the door. Password encryption is another layer of protection that makes it hard for people to read the passwords directly from the config files. This way, even if someone gets into the system, they won't be able to see the passwords easily. It's an important step in keeping the network safe from hackers and other threats.

```
%BGP-5-ADJCHANGE: neighbor 150.10.1.2 Up

Dublin#config t
Dublin#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Dublin(config)#enable secret NetDublin
Dublin(config)#end
Dublin#
%SYS-5-CONFIG_I: Configured from console by console

Dublin#end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Dublin#disable
Dublin>enable
Password:
Dublin#
```

```
Dublin#
%SYS-5-CONFIG_I: Configured from console by console

Dublin#show running
%BGP-3-NOTIFICATION: received from neighbor 150.10.1.2 4/0 (hold time expired) 0 bytes

Building configuration...

Current configuration : 1406 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Dublin
!
!
!
enable secret 5 $1$mERr$pZqRpYk/ZZ9cMajsHUz05/
!
!
!
!
!
!
ip cef
no ipv6 cef
```

Figure 1 – Password security configuration on the Dublin router showing password encryption and the encrypted enable secret.

2.2 security Banner

I set up a warning message on the Dublin router to keep it secure. Now, when someone tries to access the network, they'll see a warning first, before they can proceed.

This feature is meant to make it clear that only approved users can get into the router. It's a way to protect the device by warning people who aren't supposed to be using it that they shouldn't try to access it. This helps keep the device safe by making sure only the right people can get in.

```
*Mar 01, 12:20:49.2020: SYS-5-CONFIG_I: Configured from console by console
Dublin#exit


Dublin con0 is now available


Press RETURN to get started.


WARNING: Authorised access only.

Dublin>
```



Figure 2 – Security banner displayed when accessing the Dublin router, warning that access is restricted to authorised users.

2.3 Log timestamps

To make it easier to keep track of what's happening on the network, i turned on log timestamps on the Dublin router. This means that every system and debugging message now includes the date and time it was sent. It's really helpful for monitoring and investigating network events, because administrators can see exactly when something happened. This way, they can quickly figure out what went wrong and fix the problem.

This is useful during troubleshooting and security investigations, as events can be placed into chronological order.

```
Dublin#show running-config
Building configuration...

Current configuration : 1400 bytes
!
version 15.1
service timestamps log datetime msec
service timestamps debug datetime msec
service password-encryption
!
hostname Dublin
!
!
!
enable secret 5 $1$mERr$pZqRpYk/ZZ9cMajsHUz05/
!
!
```

Figure 3 – Dublin router configuration showing timestamps enabled for system log and debugging messages.

2.4 SSH Secure Remote Access

To set up secure remote access on the Dublin router, SSH was configured. This involved generating RSA keys and enabling SSH version 2. What makes SSH more secure than other protocols like Telnet is that it encrypts the communication between the administrator and the network device. This means that when an administrator is accessing the router remotely, the data being sent back and forth is protected from being intercepted or read by unauthorised parties. By using SSH, the risk of sensitive information being compromised is significantly reduced, making it a safer choice for remote administrative access.

I set up SSH and then tried connecting to the Dublin router from a PC. It worked - I was able to log in remotely, which meant that SSH remote access was up and running.

```
WARNING: Authorised access only.

Dublin>enable
Password:
Dublin#config t
Dublin#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Dublin(config)#ip domain-name netsec.local
Dublin(config)#crypto key generate rsa
The name for the keys will be: Dublin.netsec.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Dublin(config)#
```

Figure 4 – RSA key generation on the Dublin router for secure SSH communication.

```
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Dublin(config)#username admin secret NetDublin
*Mar 1 12:40:18.111: %SSH-5-ENABLED: SSH 1.99 has been enabled
Dublin(config)#line vty 0 4
Dublin(config-line)#login local
Dublin(config-line)#transport input ssh
Dublin(config-line)#exit
Dublin(config)#ip ssh version 2
Dublin(config)#end
Dublin#
*Mar 01, 12:53:26.5353: SYS-5-CONFIG_I: Configured from console by console
Dublin#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
Dublin#
```

Figure 5 – SSH version 2 enabled on the Dublin router.

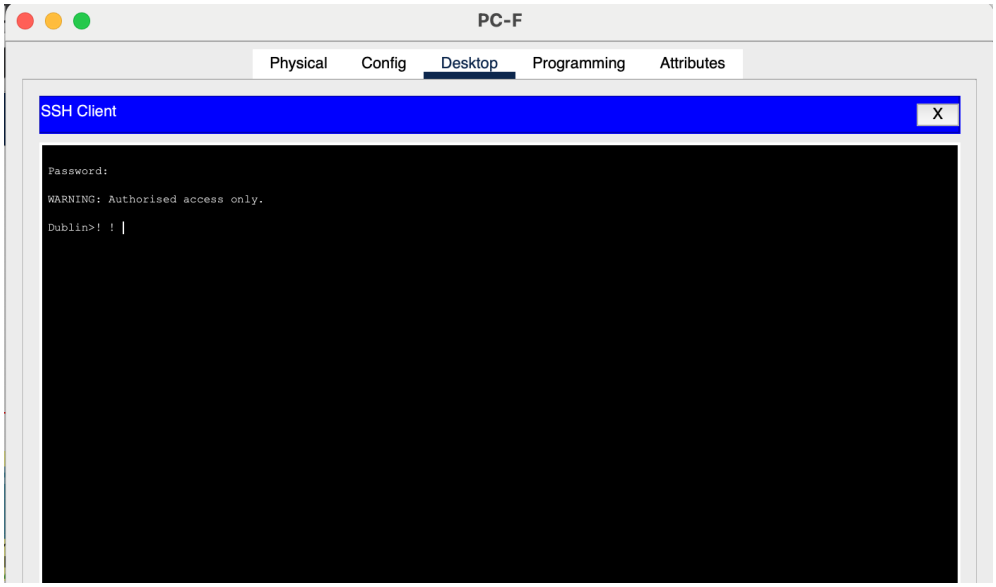


Figure 6 – Successful SSH login to the Dublin router from a network PC

3. AAA/ RADIUS Authentication

AAA authentication was implemented to improve access control to the Dublin router. A RADIUS server was used to centrally authenticate users instead of relying only on credentials stored locally on the router.

To boost security, the Dublin router was set up as a client on the AAA server, and a technician user account was created. Then, the router was configured to work with the RADIUS server. This way, user authentication can be managed from one place, and only authorised users with the right credentials can get administrative access. This makes it easier to control who can make changes to the system.

i tried to log in using SSH with the technician account to test the setup. When I was able to log in successfully, it showed that the authentication configuration was working correctly.

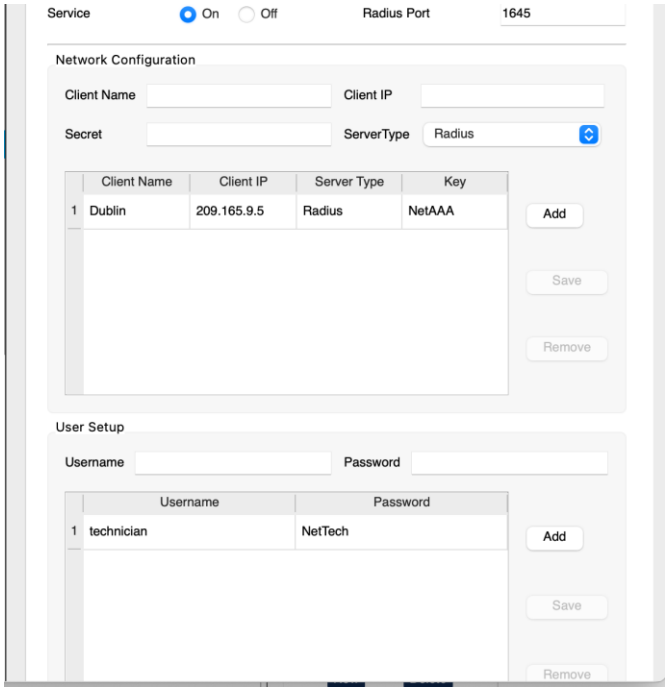


Figure 7 – RADIUS AAA server showing the Dublin router registered as an AAA client and the technician user account configured for centralised authentication

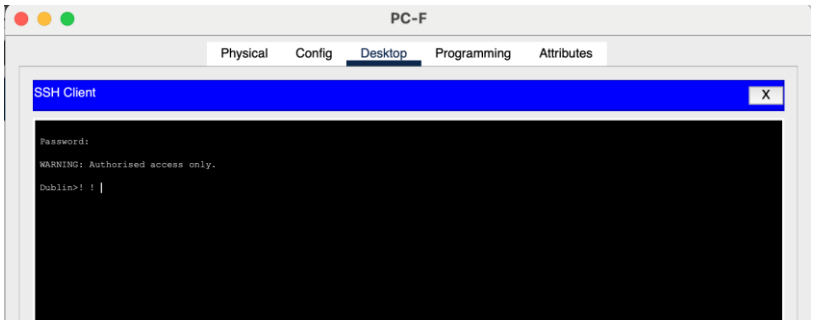


Figure 8 – Successful SSH login to the Dublin router during AAA/RADIUS authentication testing.

4. Device Monitoring – SYSLOG

SYSLOG monitoring was configured on the Dublin router so that important network events and system messages could be sent to a central server. The SYSLOG server used in the network was 192.168.3.5.

Having a central place to store logs is really helpful for keeping your network secure. This way, the person in charge can keep an eye on what's happening with all the devices on the network and use the logs to figure out what's going wrong or if someone's trying to do something they shouldn't be. It's also useful when you're trying to troubleshoot

issues with the network, because you can look back at the logs and see what was happening when the problem started.

After configuring logging, SYSLOG messages were successfully received by the server, providing evidence that the monitoring configuration was working.

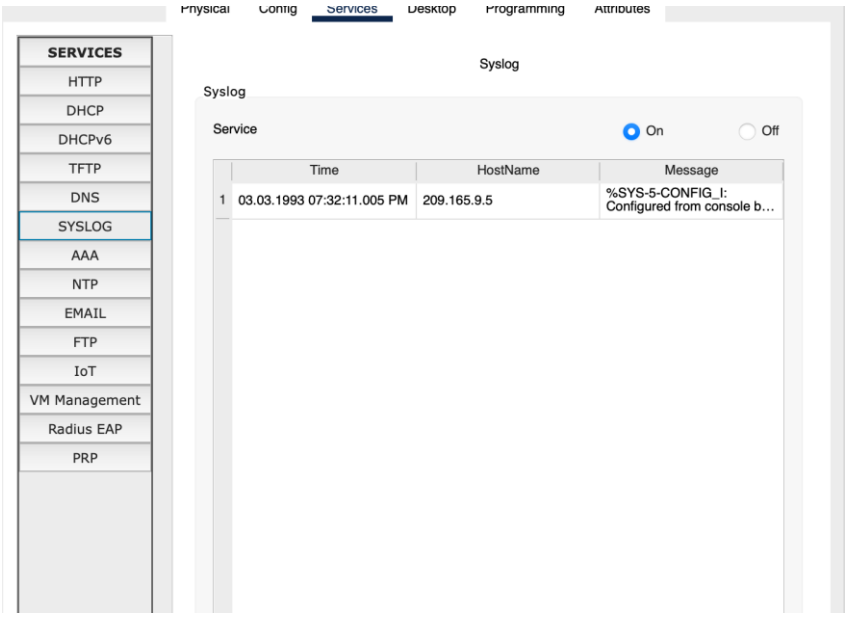


Figure 9 – SYSLOG server receiving a configuration event from the Dublin router, demonstrating centralised device monitoring.

5. Access control lists

To control the flow of traffic between different local area networks, or LANs, Access Control Lists, also known as ACLs, were set up on the router in Dublin. The goal was to stop any unauthorised communication between the Sales and Purchasing networks, while still letting legitimate traffic go through to the places it was needed. This was done to make sure that sensitive information wasn't shared between the two networks without permission.

To control traffic flow, extended ACLs were utilised, enabling filtering based on specific details like source and destination networks, as well as protocols. These ACLs were then applied to the relevant router subinterfaces, allowing unwanted traffic to be blocked at the point of entry. This approach helps prevent unnecessary traffic from passing through the router, thus improving network efficiency and security. By doing so, it ensures that only authorised traffic is allowed to flow through, reducing the risk of potential threats and maintaining a stable network environment.

I set up filtering in both directions between the Dublin local area networks. This was crucial because if I only blocked communication from Sales to Purchasing, devices in Purchasing could still start conversations with Sales. By doing it this way, I made sure that communication is restricted in both directions, which is important for security and control.

Limiting direct contact between the sales and purchasing networks helps cut down on unnecessary interactions between departments. If a device on one network gets hacked, the access control list helps stop the attacker from directly getting to devices on the other network. Meanwhile, legitimate traffic that doesn't match the blocked rules can still pass through the router. This way, the network stays safe and only authorized traffic gets through.

```
Dublin#show access-lists
Extended IP access list 100
    10 permit ip host 72.165.200.17 192.168.2.0 0.0.0.255 (4 match(es))
    20 permit udp host 72.165.200.17 host 192.168.3.2 eq domain
    25 permit tcp host 72.165.200.17 host 192.168.3.3 eq www
    27 permit tcp host 72.165.200.17 host 192.168.3.4 eq ftp
    30 deny ip 10.0.0.0 0.255.255.255 any
    40 deny ip 172.16.0.0 0.15.255.255 any
    50 deny ip 192.168.0.0 0.0.255.255 any
    60 permit ip any any (576 match(es))
Extended IP access list 110
    10 deny ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255 (8 match(es))
    20 permit ip any any
Extended IP access list 120
    10 deny ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255 (4 match(es))
    20 permit ip any any

Dublin#
```

Figure 10 - Extended ACLs configured on Dublin to restrict communication between the Sales and Purchasing LANs.

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 11– Connectivity test demonstrating that traffic from the Sales LAN to the Purchasing LAN is blocked by the ACL

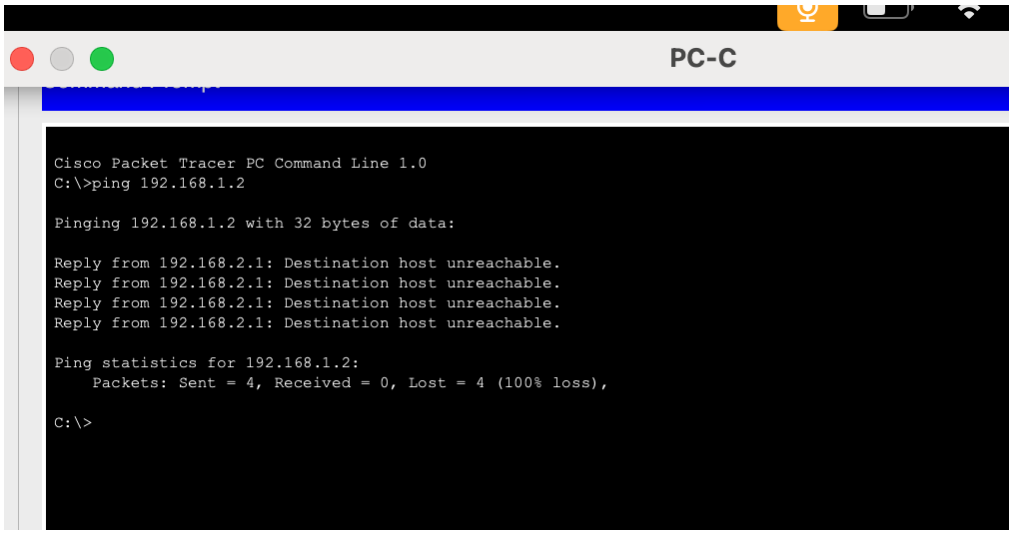


Figure 12 – Connectivity test demonstrating that traffic from the Purchasing LAN to the Sales LAN is blocked by the ACL

6. control plane security –OSPF Authentication

To make the network's routing system more secure, OSPF authentication was set up. Routers use OSPF to share information about the best routes, and if this information isn't checked, it could be a security problem. So, authentication helps ensure that only trusted routers can update the routing information, which prevents unauthorised changes that could harm the network.

To secure the communication between OSPF routers, MD5 authentication was set up. This involved configuring a shared authentication key on the interfaces that would be taking part in the process. With this in place, any OSPF routing updates that were sent had to include the correct authentication information, or they would not be accepted by the receiving router. This added an extra layer of security to the routing updates, making sure that only authorised routers could make changes to the network.

This helps keep the control plane safe from unauthorised devices that try to add fake routing information to the network.
If you don't use OSPF authentication, a router that isn't supposed to be part of the network could try to join in and send out fake routing information. But with MD5

authentication, neighbouring routers have to use the right authentication key when they talk to each other about OSPF stuff, which adds an extra layer of protection. This way, even if an unauthorised router tries to send out false information, it won't be able to because it won't have the correct key.

After turning on authentication, i checked to see if the OSPF neighbours could still talk to each other. It turned out that they could, which meant that the authentication settings were correct and that legitimate routing updates were being shared between them. This was a good sign, showing that everything was working as it should be. The routers were able to form the expected neighbour relationship, which was a key part of getting the network up and running smoothly.

```
!
!
!
!
interface GigabitEthernet0/0
 ip address 192.168.4.1 255.255.255.0
 ip nat inside
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.3.1 255.255.255.0
 ip nat inside
 duplex auto
 speed auto
!
interface Serial0/0/0
 ip address 209.165.9.6 255.255.255.252
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 NetSec123
 ip nat outside
```

Figure 13– OSPF MD5 authentication configured to protect routing updates between network routers

```
London#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
209.165.9.5       0   FULL/  -        00:00:32    209.165.9.5     Serial0/0/0
London#
```

Figure 14— OSPF neighbour verification

7. NAT/PAT - London

Port Address Translation (PAT) was configured on the London router for devices on the Admin LAN. The Admin LAN uses private addresses from the 192.168.4.0/24 network, while the London router connects to the WAN using its public-facing serial interface.

So, let's break it down - PAT is like a helper that lets multiple devices on my private network talk to the outside world using the same address as the London router's WAN interface. I used a special list to pick which internal addresses should be translated, and then I set up NAT overload on the WAN interface. This way, all my devices can share the same public address and still be able to communicate with the outside world. It's like they're all wearing the same mask, but they can still talk to each other and the world outside the network.

This helps cut down on the number of public IP addresses needed for all the devices inside a network. It also keeps the private addresses inside the network from being used directly on the public internet.

```
London#show running-config | include ip nat
 ip nat inside
 ip nat inside
 ip nat outside
ip nat inside source list 10 interface Serial0/0/0 overload
```

Figure 15 – PAT configuration on the London router showing inside/outside NAT interfaces and NAT overload

```
London#show running-config | section access-list
access-list 10 permit 192.168.4.0 0.0.0.255
```

Figure 16 – ACL 10 identifying the London Admin LAN (192.168.4.0/24) for PAT.

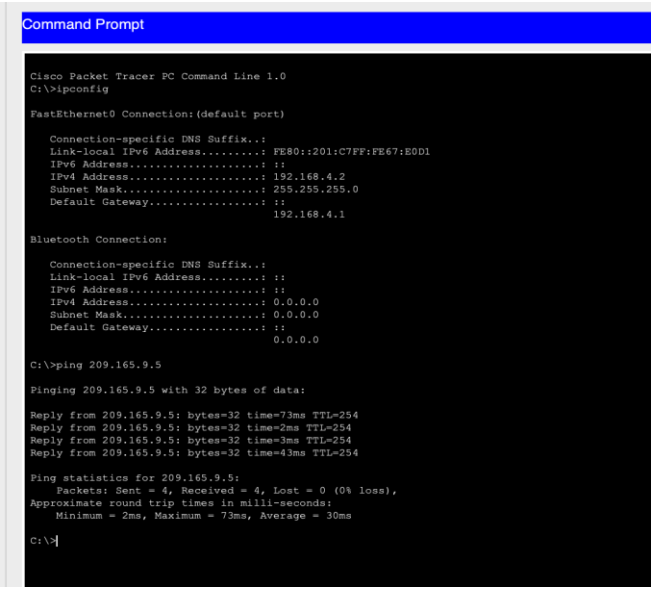


Figure 17 – Successful connectivity test from PC-F on the London Admin LAN to the London WAN interface.

7.1 static NAT limitations

The project brief also required static NAT for the servers on the London 192.168.3.0/24 network. The DNS, Web and FTP servers were identified during the implementation. However, I did not complete the individual static NAT mappings because a suitable additional public address range for mapping the internal servers was not identified during my implementation. Rather than assigning invalid public addresses, I left the server addressing unchanged

8 LAN Security

8.1 Port Security

To better protect the network, port security was put in place on the LAN switches. This was done to stop unauthorised devices from connecting to the access ports. With port security, a switch can control which devices are allowed to use a particular port, and it can also limit how many devices can connect to that port. This helps keep the network safe by only letting trusted devices access the network.

The access ports were configured as access interfaces and port security was enabled.

Port security was configured on the switch to restrict access to authorised devices. The maximum number of MAC addresses allowed on the port was set to one. This helps prevent unauthorised devices from connecting to the network through the secured switch port.

This helps protect the LAN against unauthorised devices being connected to switch ports and reduces the risk of attacks involving large numbers of spoofed MAC addresses

```
S3#show port-security interface fa0/3
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Restrict
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 0
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 0000.0000.0000:0
Security Violation Count   : 0
```

Figure 18 – Port security enabled on the switch port with a maximum of one MAC address permitted.

8.2 DHCP Snooping

To better protect the network, I also set up DHCP Snooping as a security feature at the Layer 2 level. This helps keep the network safe from fake DHCP servers by figuring out which switch interfaces are trusted and which ones are not. Basically, it distinguishes between the good guys and the bad guys, making sure only the right servers are allowed to send out DHCP messages.

DHCP Snooping was enabled on S3 for VLANs 10 and 20 to help protect the network against rogue DHCP servers. The configuration was checked using the show ip dhcp snooping command. This confirmed that DHCP Snooping was enabled for the required VLANs.

```
S3#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
10,20
Insertion of option 82 is enabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                  Trusted    Rate limit (pps)
-----------------------    -------    ----------------
GigabitEthernet0/1         yes        unlimited
S3#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
10,20
Insertion of option 82 is enabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                  Trusted    Rate limit (pps)
-----------------------    -------    ----------------
GigabitEthernet0/1         yes        unlimited
FastEthernet0/1            no         unlimited
S3#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       auto         n-802.1q       trunking      1
Fa0/2       desirable    n-802.1q       trunking      1
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/2       1-1005
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20
Fa0/2       1,10,20
Gig0/1      1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20
Fa0/2       1,10,20
Gig0/1      1,10,20
```

Figure 19 – DHCP Snooping enabled on S3 for VLANs 10 and 20

8.3 DHCP Test

I tested how DHCP works using a computer called PC-A. The computer was able to get an address from the real DHCP server, which shows that DHCP still works normally even after I changed the network security settings.

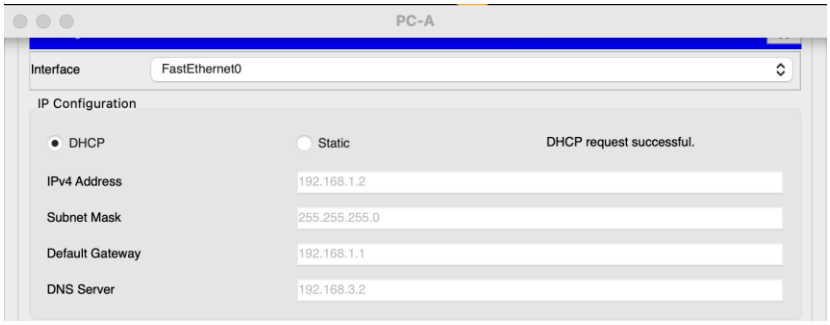


Figure 20– PC-A successfully receiving its network configuration from the legitimate DHCP server

9. WLAN Security – Beijing

9.1 WPA2 Wireless Security

To secure the wireless network at the Beijing site, I used WPA2 Personal with AES encryption. At first, the wireless security was turned off, so devices could connect without needing a password. But with WPA2 Personal and AES, I made sure only authorised devices could join the network, keeping the data safe.

To secure the wireless network, WPA2 Personal was turned on, and AES was chosen as the encryption method. A strong pre-shared key was also set up for authentication. This helps keep wireless communications safe from unauthorised access and makes it harder for others to intercept the wireless traffic being transmitted. By using a strong pre-shared key, the network is better protected against hacking and eavesdropping, giving users more confidence in the security of their wireless communications.

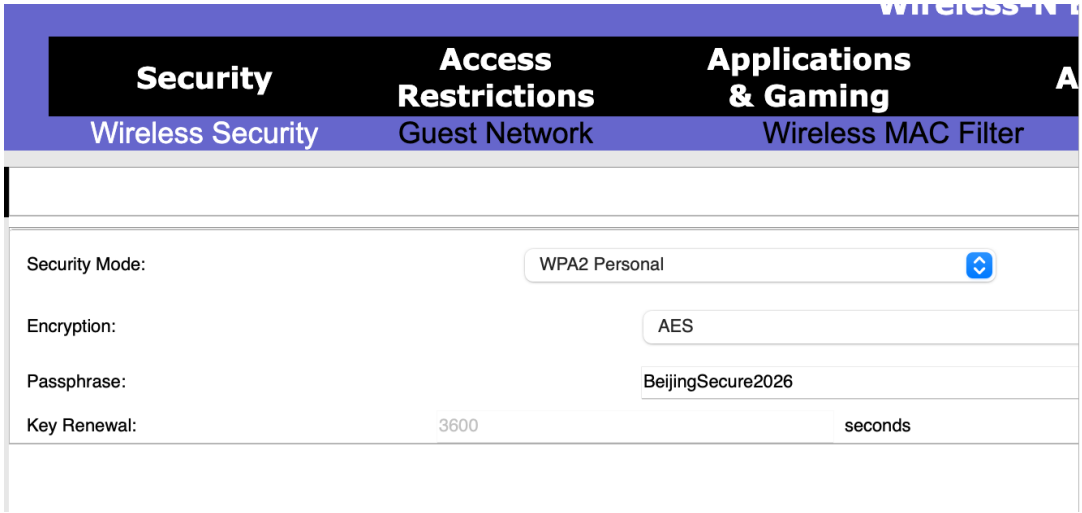


Figure 21 – Beijing wireless network secured using WPA2 Personal with AES encryption.

9.2 MAC Address Whitelisting

I also set up MAC-address filtering on the wireless router in Beijing. This filter was like a list of approved devices - only devices with MAC addresses on this list could connect to the wireless network.

I added an extra layer of security to my network by identifying the MAC address of PC-I as 00:02:16:86:30:BE and putting it on the allowed list. This works together with the WPA2 authentication to keep my network safe.

Don't rely on MAC filtering as the only way to secure your wireless network. The reason is that MAC addresses can be easily faked. But, if you use it with WPA2 and AES, it adds an extra layer of security. This means only devices you approve can connect to your network. It's like having an extra lock on the door, making it harder for unauthorised devices to get in

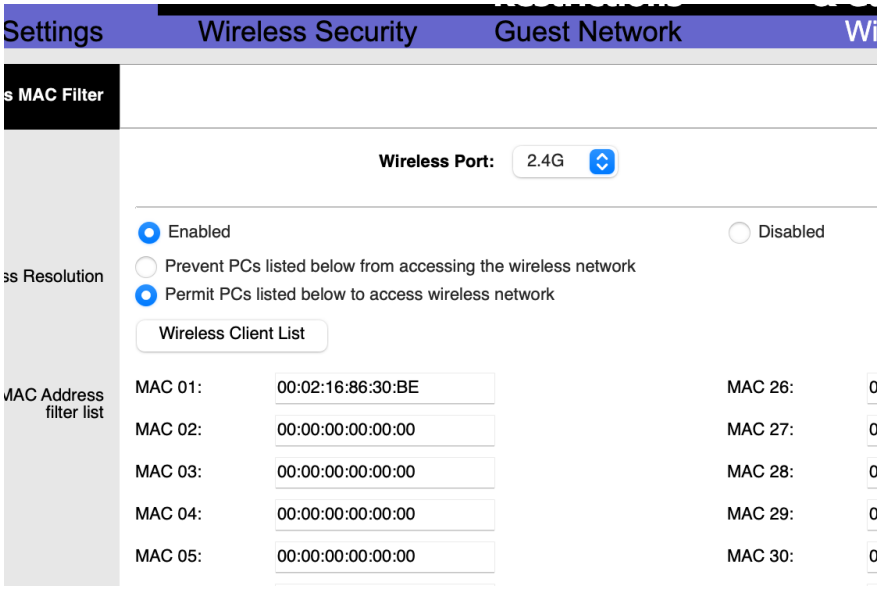


Figure 22 – MAC-address whitelisting configured on the Beijing wireless router to permit PC-I.

9.3 WLAN Testing

After putting the wireless security measures in place, i connected PC-I back to the Beijing wireless network using the WPA2 login details. PC-I was able to connect to the wireless access point without any issues since its MAC address was on the approved list.

Connectivity was then tested by pinging the Beijing default gateway at 192.168.5.1. All four ICMP packets were successfully received, resulting in 0% packet loss. This demonstrated that the authorised wireless client could still communicate successfully after the security controls were applied.

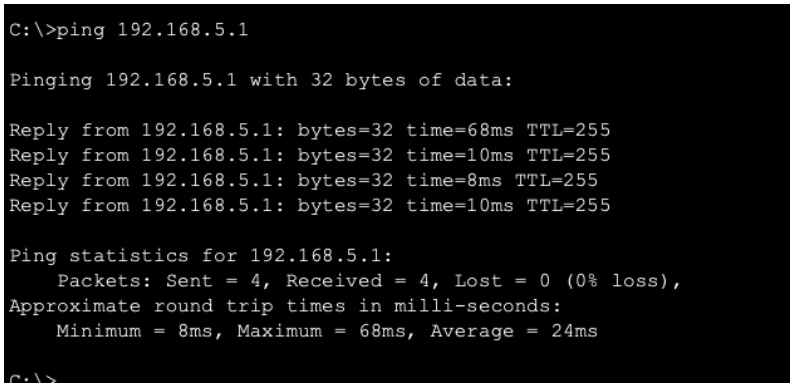


Figure 23 – Successful connectivity test from authorised wireless client PC-I to the Beijing gateway with 0% packet loss.

Testing and verification

Testing was carried out throughout the project to verify that the implemented security controls were working correctly and that legitimate network communication was still possible. Both successful and unsuccessful connectivity tests were useful because some security controls were specifically designed to block particular types of traffic.

The main tests carried out included SSH remote access, AAA authentication, ACL filtering between the Dublin LANs, DHCP operation, OSPF neighbour connectivity, London WAN connectivity and Beijing wireless connectivity.

Security Feature	Test Performed	Result
SSH	Remote SSH connection to Dublin router	Successful
AAA/RADIUS	Technician authenticated during SSH login	Successful
SYSLOG	Checked central SYSLOG server for router messages	Messages received
ACLs	Ping between Dublin Sales and Purchasing LANs	Blocked as intended
OSPF Authentication	Checked OSPF neighbour relationship	Routing remained operational
DHCP	PC-A requested addressing from DHCP server	DHCP request successful
PAT	PC-F communicated from London Admin LAN towards WAN	0% packet loss
WLAN Security	PC-I reconnected using WPA2	Successful

MAC Whitelist	Whitelisted PC-I connected to WLAN	Successful
Beijing Connectivity	PC-I pinged 192.168.5.1	0% packet loss

The testing demonstrated that several security controls were able to restrict unwanted access without preventing authorised communication. For example, the ACL tests showed that communication between the Dublin Sales and Purchasing LANs could be blocked, while other permitted network communication remained available.
SSH and AAA testing demonstrated that the Dublin router could be accessed remotely using secure authentication. The SYSLOG server also received network events, demonstrating that centralised monitoring was operating.
At the Beijing site, PC-I successfully reconnected after WPA2/AES and MAC filtering were configured. A ping to the Beijing gateway returned all four packets with 0% loss, demonstrating that the authorised wireless client retained network connectivity.

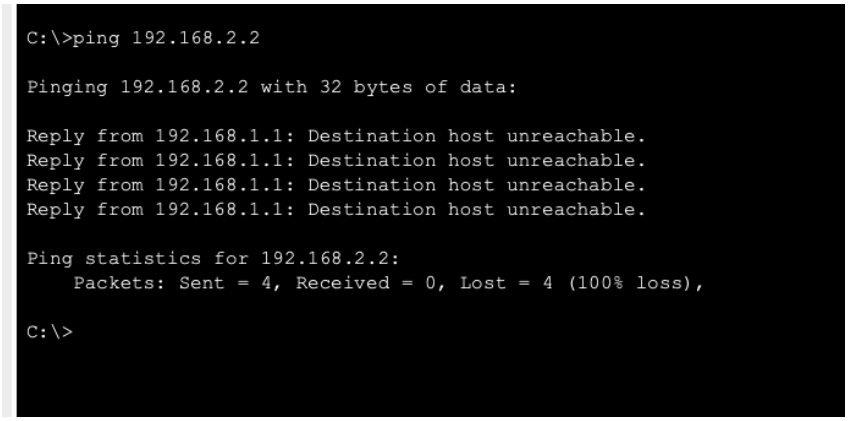


Figure 24 – Dublin ACL blocked ping

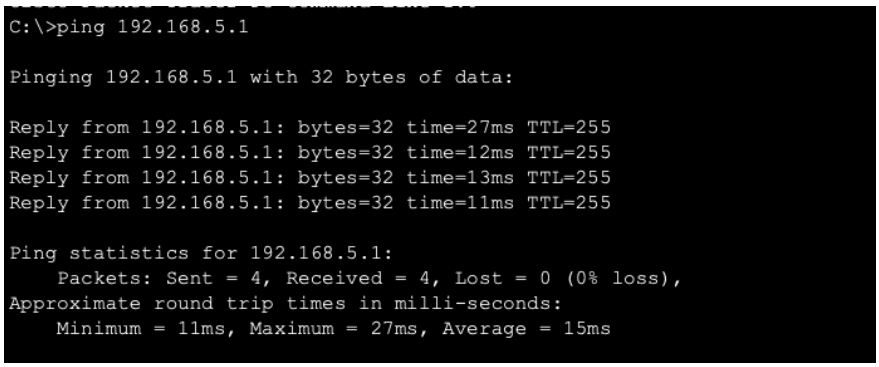


Figure 25 – Beijing WLAN successful ping

11. Limitations and security features not fully implemented

11.1 site to site vpn

One thing I didn't get to finish was setting up a secure connection between the offices in Dublin and Beijing. This connection, called a site-to-site IPsec VPN, would have encrypted the data traveling between the two locations over the public internet.

I took a closer look at the setup using the IPsec process that was explained in the module. This included things like IKE, ISAKMP, IPsec policies, and crypto maps. But when I tried to test it out on the Dublin router in Packet Tracer, I ran into some issues. It seemed that the IOS version being used didn't have the necessary ISAKMP and IPsec commands. For instance, when I tried to use the command to verify ISAKMP, it just didn't work and gave me an error message saying it was an invalid command.

To set up a secure site-to-site connection, you can use an IPsec VPN if the necessary IOS features are available. This can be done by using matching IKE policies on both Dublin and Beijing, along with pre-shared authentication.

This would help keep things private by scrambling important information that's being sent between different locations.

```
% Incomplete command.
Dublin#show crypto isakmp sa
                   ^
% Invalid input detected at '^' marker.
```

Figure 26 – Packet Tracer Dublin router showing the limitation encountered when attempting to access the required ISAKMP

11.2 Static NAT

PAT was implemented for the London Admin LAN; however, the static NAT requirement for the individual London servers was not fully completed. The internal servers were identified on the 192.168.3.0/24 network, but I did not assign static global mappings because I had not identified a suitable additional public address range for the server mappings.
When everything is set up, the right public addresses will be assigned and each necessary internal server will get a one-to-one static NAT mapping. This way, the

required server services can be accessed using predictable public addresses, making it easier to manage and connect to them.

11.3 DHCP Snooping Binding Table

DHCP Snooping was turned on for VLANs 10 and 20 on S3 and it worked. The interface that connects to the legitimate DHCP server was set up as a trusted one. I also tested DHCP with PC-A and it worked fine.

But when I tested it in Packet Tracer, the DHCP Snooping binding table didn't fill up like it was supposed to. So, I couldn't say for sure that the dynamic DHCP Snooping bindings were working right. I did check the configuration using the "show ip dhcp snooping" command, and that looked good. And I also tested the regular DHCP setup from PC-A, and that worked fine too.

This limitation prevented me from demonstrating additional security tests that rely on DHCP Snooping bindings.

12. Conclusion

This project was all about making a multi-site network secure. The network had sites in Dublin, London, and Beijing, and **I** used Cisco Packet Tracer to work on it. **My** main goal was to make the network more secure without disrupting the connections that legitimate users and services need. **I** wanted to find a balance between security and connectivity, so that the people who are supposed to be using the network can still get in and do their work, while keeping the bad guys out.

To better protect the network, several security measures were put in place during the project. For instance, devices were made more secure by adding password protection, encrypting passwords, displaying security banners, logging timestamps, and allowing secure remote access via SSH. A centralized system for authentication was also set up using AAA with RADIUS, which made it easier to manage user access. Additionally, SYSLOG was configured to monitor network events from a central location, providing a clearer picture of what's happening on the network. To control traffic flow, Access Control Lists were used to limit communication between the Dublin Sales and Purchasing networks, ensuring that sensitive data wasn't accessible to unauthorized parties. Furthermore, OSPF authentication was implemented to secure routing updates, preventing malicious actors from manipulating network traffic. These security controls work together to provide a robust defence against potential threats.

Layer 2 security was improved using port security and DHCP Snooping. At the Beijing site, the wireless network was secured using WPA2 Personal with AES encryption and MAC-address whitelisting. PAT was also configured on the London router to allow devices on the private Admin LAN to communicate through the WAN interface.

I made sure **my** security controls were working properly by testing them throughout the setup process. **My** tests showed that several key features were functioning as expected. For instance, **I** was able to successfully authenticate using SSH, and **my** SYSLOG messages were being transmitted correctly. **I** also confirmed that **my** DHCP addressing was working, and that **my** OSPF connectivity was established. Additionally, **I** tested and verified **my** London WAN connectivity, as well as **my** Beijing wireless connectivity. Furthermore, **my** ACL testing demonstrated that **I** was able to effectively block traffic between the Dublin LANs that was intended to be restricted. Overall, **my** testing gave **me** confidence that **my** security controls were in place and working as intended.

Some requirements were not fully completed, particularly the site-to-site IPsec VPN and static NAT for the London servers. These have been discussed as limitations of the implementation and could be completed in future development of the network. Despite these limitations, the project demonstrates the use of several different network security techniques across the network and shows how multiple security controls can be combined to provide a more secure environment.